V - Bio molecules-Nutrient test

Exercise: 28

Test for reducing sugar – Benedict reagent test

Aim:

To detect the presence of reducing sugar.

Basic Principle:

1. Aldoses and Ketoses are reducing sugars. Glucose is the reducing sugar and sucrose is the non-reducing sugar.
2. When reducing sugar is heated with an alkaline solution of Copper (II) sulphate (Benedict's solution) reduces Cu^{2++} into Cu^{+} forming brick red precipitate of Copper (I) oxide.

Requirements:

Test tube, test tube stand, test tube holder, Samples for test- Fruit juices of apples/ banana/ leaves of onion, sugar cane extract, milk etc., Benedict's solution, spirit lamp, water bath.

Procedure:

1. Take 1 ml of sample solution in a clean test tube
2. Add 1 ml of Benedict's solution
3. Keep the test tube in the boiling water bath.
4. Appearance of brick red colour depends on concentration of reducing sugar.

Table:

Procedure	Observation	Inference
1ml of sample solution + 1ml of Benedict's solution, Heated	Appearance of brick red colour	Reducing sugar is present (Glucose is the reducing sugar)

Exercise: 29

Test for starch – Iodine test

Aim:

To detect the presence of starch in the given sample solution.

Basic Principle:

1. Starch is the storage polysaccharide of plants.
2. It consist of two component a. amylose (linear, unbranched polymer, soluble in water) b. amylopectin (a branched polymer)
3. Amylose portion of starch react with Iodine (Potassium iodide) produces deep blue-black colour.

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